

OpenStax Calculus Volume 1

Section 4.5 Exercises: Derivatives and the Shape of a Graph

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1. If c is a critical point of $f(x)$, when is there no local maximum or minimum at c ? Explain.
2. For the function $y = x^3$, is $x = 0$ both an inflection point and a local maximum/minimum?
3. For the function $y = x^3$, is $x = 0$ an inflection point?
4. Is it possible for a point c to be both an inflection point and a local extremum of a twice differentiable function?
5. Why do you need continuity for the first derivative test? Come up with an example.
6. Explain whether a concave-down function has to cross $y = 0$ for some value of x .
7. Explain whether a polynomial of degree 2 can have an inflection point.

For the following exercises, analyze the graphs of f' , then list all intervals where f is increasing or decreasing.

8. *[Figure: The function $f'(x)$ is graphed. The function starts negative and crosses the x -axis at $(-2, 0)$. Then it continues increasing a little before decreasing and crossing the x -axis at $(-1, 0)$. It achieves a local minimum at $(1, -6)$ before increasing and crossing the x -axis at $(2, 0)$.]*
9. *[Figure: The function $f'(x)$ is graphed. The function starts negative and crosses the x -axis at $(-2, 0)$. Then it continues increasing a little before decreasing and touching the x -axis at $(-1, 0)$. It then increases a little before decreasing and crossing the x -axis at the origin. The function then decreases to a local minimum before increasing, crossing the x -axis at $(1, 0)$, and continuing to increase.]*
10. *[Figure: The function $f'(x)$ is graphed. The function starts negative and touches the x -axis at the origin. Then it decreases a little before increasing to cross the x -axis at $(1, 0)$ and continuing to increase.]*
11. *[Figure: The function $f'(x)$ is graphed. The function starts positive and decreases to touch the x -axis at $(-1, 0)$. Then it increases to $(0, 4.5)$ before decreasing to touch the x -axis at $(1, 0)$. Then the function increases.]*
12. *[Figure: The function $f'(x)$ is graphed. The function starts at $(-2, 0)$, decreases to $(-1.5, -1.5)$, increases to $(-1, 0)$, and continues increasing before decreasing to the origin. Then the other*

side is symmetric: that is, the function increases and then decreases to pass through $(1, 0)$. It continues decreasing to $(1.5, -1.5)$, and then increase to $(2, 0)$.]

For the following exercises, analyze the graphs of f' , then list all intervals where

- (a) f is increasing and decreasing and
- (b) the minima and maxima are located.

13. [Figure: The function $f'(x)$ is graphed. The function starts at $(-2, 0)$, decreases for a little and then increases to $(-1, 0)$, continues increasing before decreasing to the origin, at which point it increases.]
14. [Figure: The function $f'(x)$ is graphed. The function starts at $(-2, 0)$, increases and then decreases to $(-1, 0)$, decreases and then increases to an inflection point at the origin. Then the function increases and decreases to cross $(1, 0)$. It continues decreasing and then increases to $(2, 0)$.]
15. [Figure: The function $f'(x)$ is graphed from $x = -2$ to $x = 2$. It starts near zero at $x = -2$, but then increases rapidly and remains positive for the entire length of the graph.]
16. [Figure: The function $f'(x)$ is graphed. The function starts negative and crosses the x -axis at the origin, which is an inflection point. Then it continues increasing.]
17. [Figure: The function $f'(x)$ is graphed. The function starts negative and crosses the x -axis at $(-1, 0)$. Then it continues increasing a little before decreasing and touching the x -axis at the origin. It increases again and then decreases to $(1, 0)$. Then it increases.]

For the following exercises, analyze the graphs of f' , then list all inflection points and intervals where f is concave up and concave down.

18. [Figure: The function $f'(x)$ is graphed. The function is linear and starts negative. It crosses the x -axis at the origin.]
19. [Figure: The function $f'(x)$ is graphed. It is an upward-facing parabola with 0 as its local minimum.]
20. [Figure: The function $f'(x)$ is graphed. The function resembles the graph of x^3 : that is, it starts negative and crosses the x -axis at the origin. Then it continues increasing.]
21. [Figure: The function $f'(x)$ is graphed. The function starts negative and crosses the x -axis at $(-0.5, 0)$. Then it continues increasing to $(0, 1.5)$ before decreasing and touching the x -axis at $(1, 0)$. It then increases.]
22. [Figure: The function $f'(x)$ is graphed. The function starts negative and crosses the x -axis at $(-1, 0)$. Then it continues increasing to a local maximum at $(0, 1)$, at which point it decreases and touches the x -axis at $(1, 0)$. It then increases.]

For the following exercises, draw a graph that satisfies the given specifications for the domain $x \in [-3, 3]$. The function does not have to be continuous or differentiable.

23. $f(x) > 0, f'(x) > 0$ over $x > 1, -3 < x < 0, f'(x) = 0$ over $0 < x < 1$
24. $f'(x) > 0$ over $x > 2, -3 < x < -1, f'(x) < 0$ over $-1 < x < 2, f''(x) < 0$ for all x

25. $f''(x) < 0$ over $-1 < x < 1$, $f''(x) > 0$, $-3 < x < -1$, $1 < x < 3$, local maximum at $x = 0$, local minima at $x = \pm 2$
26. There is a local maximum at $x = 2$, local minimum at $x = 1$, and the graph is neither concave up nor concave down.
27. There are local maxima at $x = \pm 1$, the function is concave up for all x , and the function remains positive for all x .

For the following exercises, determine

- (a) intervals where f is increasing or decreasing and
- (b) local minima and maxima of f .

If needed, use a calculator to graph the functions, but show your work.

28. $f(x) = \sin x + \sin^3 x$ over $-\pi < x < \pi$

29. $f(x) = x^2 + \cos x$

For the following exercises, determine a. intervals where f is concave up or concave down, and b. the inflection points of f .

30. $f(x) = x^3 - 4x^2 + x + 2$

For the following exercises, determine

- (a) intervals where f is increasing or decreasing,
- (b) local minima and maxima of f ,
- (c) intervals where f is concave up and concave down, and
- (d) the inflection points of f .

31. $f(x) = x^2 - 6x$

32. $f(x) = x^3 - 6x^2$

33. $f(x) = x^4 - 6x^3$

34. $f(x) = x^{11} - 6x^{10}$

35. $f(x) = x + x^2 - x^3$

36. $f(x) = x^2 + x + 1$

37. $f(x) = x^3 + x^4$

For the following exercises, determine

- (a) intervals where f is increasing or decreasing,
- (b) local minima and maxima of f ,
- (c) intervals where f is concave up and concave down, and
- (d) the inflection points of f . Sketch the curve, then use a calculator to compare your answer. If you cannot determine the exact answer analytically, use a calculator.

38. [T] $f(x) = \sin(\pi x) - \cos(\pi x)$ over $x = [-1, 1]$
39. [T] $f(x) = x + \sin(2x)$ over $x = [-\frac{\pi}{2}, \frac{\pi}{2}]$
40. [T] $f(x) = \sin x + \tan x$ over $(-\frac{\pi}{2}, \frac{\pi}{2})$
41. [T] $f(x) = (x - 2)^2(x - 4)^2$
42. [T] $f(x) = \frac{1}{1-x}, x \neq 1$
43. [T] $f(x) = \frac{\sin x}{x}$ over $x = [-2\pi, 0) \cup (0, 2\pi]$
44. $f(x) = \sin(x)e^x$ over $x = [-\pi, \pi]$
45. $f(x) = (\ln x)\sqrt{x}, x > 0$
46. $f(x) = \frac{1}{4}\sqrt{x} + \frac{1}{x}, x > 0$
47. $f(x) = \frac{e^x}{x}, x \neq 0$

For the following exercises, interpret the sentences in terms of f , f' , and f'' .

48. The population is growing more slowly. Here f is the population.
49. A bike accelerates faster, but a car goes faster. Here f = Bike's position minus Car's position.
50. The airplane lands smoothly. Here f is the plane's altitude.
51. A company tracks labor costs over the course of a year. On July 1, costs are at their peak. Here f represents the cost of labor.
52. The economy is picking up speed. Here f is a measure of the economy, such as GDP.

For the following exercises, consider a third-degree polynomial $f(x)$, which has the properties $f'(1) = 0, f'(3) = 0$ (not applicable to Exercises 249 and 250). Determine whether the following statements are true or false. Justify your answer.

53. $f(x) = 0$ for some $1 \leq x \leq 3$
54. $f''(x) = 0$ for some $1 \leq x \leq 3$
55. There is no absolute maximum at $x = 3$
56. If $f(x)$ has three roots, then it has 1 inflection point.
57. If $f(x)$ has one inflection point, then it has three real roots.